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BANGALORE · INDIA

BUNDLEWISE: A RETAIL ANALYTICS TOOL FOR SMART BUNDLING AND PROFIT-DRIVEN PRICING

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Course:

MDS 481 – Project

Mission

Christ University is a nurturing ground for an individual's holistic development to make effective contribution to the society in a dynamic environment

Vision

Excellence and Service

Core Values

Faith in God | Moral Uprightness
Love of Fellow Beings | Social Responsibility
Pursuit of Excellence

INTRODUCTION TO RETAIL ANALYTICS

- Retail analytics transforms raw sales data into actionable insights.
- Enables retailers to:
 - Discover frequently bought together products.
 - Analyze customer purchase patterns over time.
 - Make smarter decisions on pricing, bundling, and promotions.
- Traditional analytics tools are often complex and costly and require technical expertise.
- BundleWise addresses this gap by offering an easy-to-use, web-based solution tailored for small and medium retailers.

PROBLEM STATEMENT (RETAILERS' ISSUES)

- Small and medium retailers struggle to make effective use of their sales data.
- Their records are often scattered across spreadsheets or POS exports, making it difficult to analyze consistently.
- Without analytics expertise, they cannot apply machine learning or statistical models to uncover patterns.
- Existing tools provide only basic sales summaries and fail to highlight important insights such as product bundles, purchase sequences, or pricing opportunities.
- As a result, retailers miss out on cross-selling, upselling, and optimized pricing strategies, which directly reduces their profitability and competitiveness.

INTRODUCTION TO BUNDLEWISE (OUR SOLUTION)

BundleWise is a web-based retail analytics tool designed to solve these issues:

- **Unified Data Handling:** Accepts sales data directly in CSV/XLSX/PDF format and preprocesses automatically.
- **Smart Bundling & Sequences:** Uses Eclat for Market Basket Analysis and PrefixSpan for Sequential Pattern Mining.
- **Profit-Driven Pricing:** Applies optimization models to recommend effective bundle pricing.
- **Easy-to-Use Dashboard:** Visualizes insights clearly, so even non-technical users can act on data.

With BundleWise, retailers can transform raw sales data into actionable insights for bundling, sequencing, and pricing — boosting sales and profitability.

MOTIVATION

The retail industry is becoming more competitive, with customers expecting **better deals, attractive bundles, and personalized experiences**. Small and medium retailers understand the importance of using data to make decisions, but they often lack access to data experts or expensive tools.

This creates the need for:

- **Smart Bundling** – finding products that are often bought together (e.g., Chips + Coke).
- **Profit-Driven Pricing** – setting prices that improve sales and maintain profits.
- **Simple Dashboards** – helping retailers make quick decisions without technical skills.

Our motivation is to build a solution that makes **data analytics easy, affordable, and useful**, so retailers can use their sales data effectively and stay competitive.

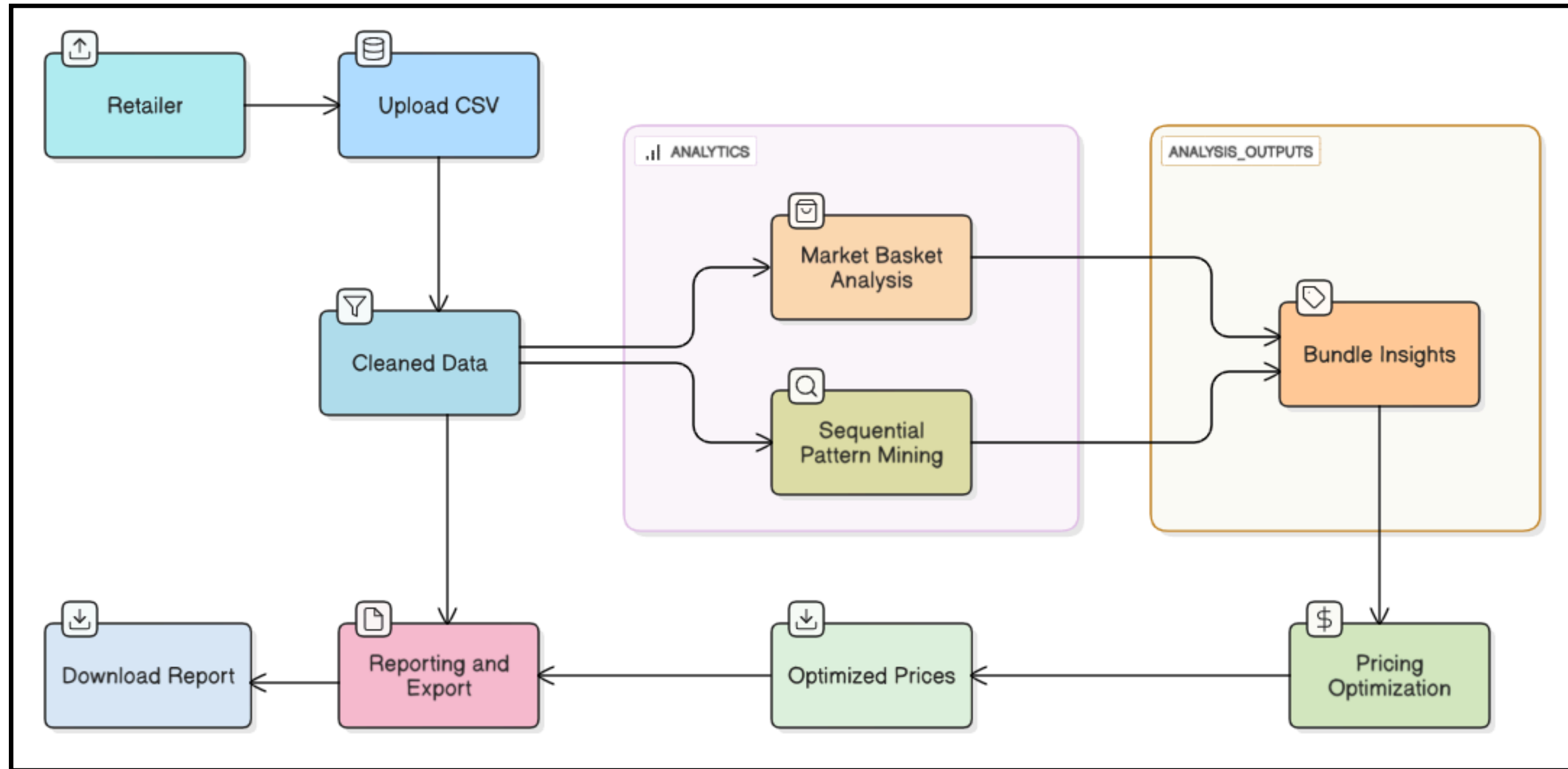
OBJECTIVES

- The goal of BundleWise is to provide a comprehensive yet user-friendly retail analytics platform. Our objectives are:
 - **Identify Product Bundles** using Market Basket Analysis (Eclat).
 - **Discover Sequential Purchase Patterns** using PrefixSpan.
 - **Recommend Pricing Strategies** that maximize profitability.
 - **Provide an Interactive Dashboard** that visualizes insights clearly for non-technical users.
 - **Enable report export and systematic testing** to ensure reliability

FEATURES OF BUNDLEWISE

- **Easy Data Upload** – accepts sales data in CSV/XLSX/PDF format.
- **Automated Preprocessing** – cleans, validates, and prepares data for analysis.
- **Market Basket Analysis (Eclat)** – finds products frequently bought together.
- **Sequential Pattern Mining (PrefixSpan)** – discovers the order in which products are purchased.
- **Bundle Recommendation** – suggests meaningful product bundles for cross-selling.
- **Pricing Optimization** – recommends competitive prices for bundles to maximize profits.
- **Interactive Dashboard** – simple visual interface for non-technical users.
- **Report Export** – generate downloadable reports in PDF and pre-processed data

WORKFLOW OF BUNDLEWISE



TECHNOLOGIES AND TOOLS USED

Component	Technology/Tool	Purpose
Programming Language	Python	Core language for backend logic and analytics
Frontend (UI)	HTML, CSS, JavaScript	User interface for data upload and interactive dashboard
Backend (APIs)	Flask	Handles requests and connects frontend with analytics engine
Data Handling	Pandas, NumPy	Data ingestion, cleaning, transformation, and manipulation
Market Basket Analysis	mlxtend (Eclat implementation)	Discovering frequently co-purchased items within transactions
Sequential Pattern Mining	mlxtend, SPMF (PrefixSpan)	Identifying product purchase sequences across customer visits
Pricing Optimization	SciPy	Recommending bundle prices to maximize sales and profit
Visualization	Plotly, Matplotlib	Creating clear and interactive data visualizations
Storage	CSV/XLSX/PDF files (local ingestion, extendable to DB)	Storing and processing retailer sales data

MODULES IN BUNDLEWISE

BundleWise is divided into five main modules:

1. **Data Ingestion Module** – Uploads sales data (CSV/XLSX/PDF), validates it, and preprocesses into a clean format.
2. **Sequential Pattern Mining (SPM)** – Implements PrefixSpan to discover purchase sequences.
3. **Market Basket Analysis (MBA)** – Implements Eclat algorithm to identify frequently bought-together products.
4. **Price Optimization Module** – Applies optimization models (using SciPy) to recommend effective bundle pricing strategies.
5. **Dashboard & Visualization** – Combines results, integrates pricing strategies, and displays insights in charts for easy interpretation.

WHY MBA AND SPM FOR BUNDLING?

Market Basket Analysis (Eclat):

- Ideal for identifying static bundles – items frequently purchased together in the same transaction.
- Example: Chips + Coke.

Sequential Pattern Mining (PrefixSpan):

- Designed for discovering temporal bundles – products bought in a sequence across multiple visits.
- Example: Shoes → Socks → Polish.

Why use both?

- MBA highlights co-occurrence patterns (what customers buy together).
- SPM uncovers purchase flow patterns (what customers buy next).
- Combined, they enable retailers to create:
 - Combo Offers → Boost immediate sales.
 - Cross-Visit Promotions → Strengthen long-term customer engagement.

TYPES OF BUNDLES IMPLEMENTED

BundleWise generates two types of actionable bundles, each designed to benefit retailers in different ways:

1. Complementary Collection Bundles (via MBA – Eclat)

- Example: Chips + Coke bought together in the same transaction.
- Retailer Benefit:
 - Can offer these as a combo pack at a small discount.
 - Increases basket value per visit.
 - Improves shelf placement decisions (place chips and Coke close together).

2. Buy Now, Save Next Bundles (via SPM – PrefixSpan)

- Example: Shoes → Socks → Polish purchased across multiple visits.
- Retailer Benefit:
 - Enables “Buy now, save next” offers (e.g., buy shoes today, get discount on socks next time).
 - Encourages repeat purchases and customer loyalty.
 - Builds long-term revenue instead of only single-visit sales.

MODULE 1: DATA INGESTION & PREPROCESSING

Intelligent Data Onboarding & Preparation:

- **Multi-Format Ingestion:** Seamlessly processes .csv, .xlsx, and tabular .pdf files, providing flexibility for various data sources.
- **Automated Schema Mapping:** Intelligently renames common column variations (e.g., invoiceno, InvoiceNo) to a standardized internal schema (Transaction ID), reducing the need for manual file editing.
- **Robust Schema Validation:** Guarantees data integrity by verifying the presence of all critical columns required for analysis (Transaction ID, Product ID, Customer ID, etc.) before processing.
- **Data Type Coercion:** Automatically converts columns to their correct data types, ensuring dates are treated as datetime objects and numerical fields like Quantity and Price are correctly formatted.

MODULE 1: DATA INGESTION & PREPROCESSING

Comprehensive Data Cleansing:

- **Missing Value Handling:** Automatically discards rows missing critical identifiers to ensure analysis quality.
- **Duplicate Record Removal:** Identifies and purges identical transaction records to prevent inflated metrics.
- **Business Logic Validation:** Filters out invalid records, such as transactions with zero or negative quantities/prices.

Transparent Processing: If the cleaning process removes all data rows (due to poor source file quality), the system provides a clear, actionable error message to the user instead of failing silently.

MODULE 2: SEQUENTIAL PATTERN MINING

Customer Journey & Behavior Analysis:

- **Sequence Generation Engine:** Transforms raw transactional data into structured customer journey sequences, forming the foundation for temporal analysis.
- **Efficient Pattern Discovery:** Implements the high-performance PrefixSpan algorithm to discover frequent purchase sequences without the performance drawbacks of older algorithms.
- **Actionable Insight Generation:** The results are presented in a detailed table showing the pattern, its statistical support, and metrics for confidence, allowing for easy identification of opportunities for timed marketing and promotions.
- **Big Data Safeguards:** For datasets with a very large number of customers, the system automatically performs intelligent customer-based sampling to deliver rapid insights while preserving statistical relevance.

MODULE 3: MARKET BASKET ANALYSIS (BUNDLE RECOMMENDATION)

- Market Basket Analysis identifies frequent itemsets – groups of products often bought together in the same transaction.
- In BundleWise, we use the **Eclat algorithm**, which is known for being fast, simple, and efficient compared to other MBA algorithms.
- Compared to Apriori:
 - Apriori repeatedly scans the dataset → slower on larger data.
 - Eclat uses tidlist intersections → fewer scans, faster execution.
- Compared to FP-Growth:
 - FP-Growth requires building complex tree structures.
 - Eclat is simpler to implement and easier to interpret.
- Eclat is the best balance of speed, simplicity, and interpretability for small and medium retailers.

MODULE 3: MARKET BASKET ANALYSIS

(BUNDLE RECOMMENDATION)

Eclat (Equivalence Class Transformation) is a frequent itemset mining algorithm. In BundleWise, it's used to discover static bundles (products often bought together in the same basket).

Step 1 – Represent transactions as TID sets

- Each product is associated with the list of transaction IDs (TIDs) where it appears.

Transaction ID	Items Bought	Transaction ID
T1	Chips, Coke, Bread	T1
T2	Chips, Coke	T2
T3	Coke, Pizza	T3
T4	Chips, Chocolate	T4

Convert into tidlists:

Chips $\rightarrow$ {T1, T2, T4}

Coke $\rightarrow$ {T1, T2, T3}

Bread $\rightarrow$ {T1}

Chocolate $\rightarrow$ {T4}

MODULE 3: MARKET BASKET ANALYSIS (BUNDLE RECOMMENDATION)

Step 2 – Intersection of tidlists

- To check if two products form a frequent itemset, intersect their TID lists.
- **Example:**
 - $\{\text{Chips}\} \cap \{\text{Coke}\} = \{\text{T1}, \text{T2}\} \rightarrow$ Appears in 2 transactions.
 - $\text{Support} = 2 / 4 = 50\%$.

Step 3 – Generate frequent itemsets

- If $\text{support} \geq \text{threshold}$, the itemset is considered frequent.
- **Example:**
 - $\{\text{Chips}, \text{Coke}\}$ passes min support $\rightarrow$ frequent bundle.
 - $\{\text{Chips}, \text{Bread}\}$ appears only once $\rightarrow$ not frequent.

MODULE 3: MARKET BASKET ANALYSIS

(BUNDLE RECOMMENDATION)

Association Rules (with Support, Confidence, Lift)

A bundle is recommended only if $\text{Support} \geq 0.1\%$, $\text{Confidence} \geq 40\%$, and $\text{Lift} \geq 1.2$.

(e.g., “Chips $\rightarrow$ Coke: $\text{Support} = 45\%$, $\text{Confidence} = 80\%$, $\text{Lift} = 1.5 \rightarrow$ Recommended”)

Metric	Definition	Threshold in BundleWise	Interpretation
Support	How often items appear together in all transactions	$\geq 0.1\%$	Frequency of co-occurrence
Confidence	Probability that if a customer buys A, they will also buy B	$\geq 40\%$	Predictive strength of the rule
Lift	How much more likely items are bought together vs. random chance	≥ 1.2	Strength of association (>1 = positive, $=1$ none, <1 negative)

BUNDLE RECOMMENDATION VALIDATION

1. Statistical Validation

- Minimum thresholds: Support $\geq 0.1\%$, Confidence $\geq 40\%$, Lift ≥ 1.2 .
- Bidirectional check prevents duplicates ($A \rightarrow B$ vs $B \rightarrow A$).
- Ensures bundles are backed by sufficient data volume.

2. Business Logic Validation

- Price compatibility (second item not $> 2x$ base item).
- Co-occurrence $\geq 15\%$ of the time.
- Only valid if products are realistically sold together.

3. Temporal Validation

- Sequential validation for SPM bundles.
- Confirms items are not just coincidental but part of a real buying pattern.

BundleWise only recommends bundles that are statistically strong, business-viable, and customer-proven.

MODULE 4: PRICING OPTIMIZATION FOR BUNDLES

- Different bundle types (MBA vs SPM) require different pricing approaches.
- **Goal:** Maximize profitability while ensuring customer value.

1. Complementary Collection Bundles (via MBA – Eclat)

- **Example:** Blue Felt Easter Egg Basket + Cream Felt Easter Egg Basket
- **Pricing Strategy:**
 - Graduated Discount → “Buy one basket, get the second at 50% off.”
 - **Example:** Buy Blue Felt Easter Egg Basket → receive 50% off offer on Cream Felt Easter Egg Basket.
- **Why This Strategy?**
 - These products have a very high co-occurrence in the same basket.
 - Customers often need multiple units, especially during festive seasons.
- **Retailer Benefits:**
 - Boosts basket value by encouraging multi-item purchases.
 - Improves inventory turnover for seasonal or fast-moving goods.
 - Drives impulse buying with attractive combo deals.

MODULE 4: PRICING OPTIMIZATION FOR BUNDLES

2. Buy Now, Save Next Bundles (via SPM – PrefixSpan)

- **Example:** White Hanging Heart T-Light Holder → Regency Cakestand
- **Pricing Strategy:**
 - Optimized Sequential Discount → 20–30% off the second product, based on purchase confidence level.
 - **Example:** Buy a T-Light Holder today → receive a targeted offer for a Cakestand on the next visit.
- **Why This Strategy?**
 - Sequential buying patterns indicate future purchase intent.
 - Targeted discounts ensure customers return to complete the sequence.
- **Retailer Benefits:**
 - Encourages repeat visits, improving customer retention.
 - Strengthens loyalty through personalized offers.
 - Expands revenue by connecting different product categories (home décor → serveware).

MODULE 4: PRICING OPTIMIZATION FOR BUNDLES

Key Retailer Impact

Bundle Type	Strategy	Retailer Gains
Complementary Collection (MBA)	Graduated discount (50% off 2nd item)	Larger basket size, faster inventory turnover, stronger promotions
Buy Now, Save Next (SPM)	Sequential discount (20–30% off next item)	More repeat visits, higher customer lifetime value, cross-category sales

BundleWise ensures retailers don’t just see what customers are buying — they also get clear, actionable pricing strategies that translate insights into measurable business growth

DASHBOARD & USER EXPERIENCE

- **Centralized Analytics Hub**
 - **Interactive dashboard with four key views:**
 - **Summary** → Revenue, transactions, AOV, sales trends
 - **Product Performance** → Top products, price-volume analysis, revenue concentration
 - **Customer Insights** → Spending patterns, repeat purchases, customer segmentation
 - **Advanced Analytics** → Cross-sell heatmaps, seasonal trends, profitability
- **User-Centric Design**
 - Intuitive navigation with tooltips and hover info panels
 - Visual KPIs and charts designed for non-technical users
 - Clean interface with clear validation/error messages
- **Actionable Insights**
 - AI-powered bundle & pricing recommendations displayed with savings and revenue impact
 - Exportable reports (PDF/Excel) mirror dashboard visuals for offline decision-making
- **Retailer Benefits**
 - Quick, data-driven decisions without technical expertise
 - Engaging experience that bridges raw data with practical retail strategies

DASHBOARD - SUMMARY TAB

1. Revenue Distribution (Donut/Pie Chart)

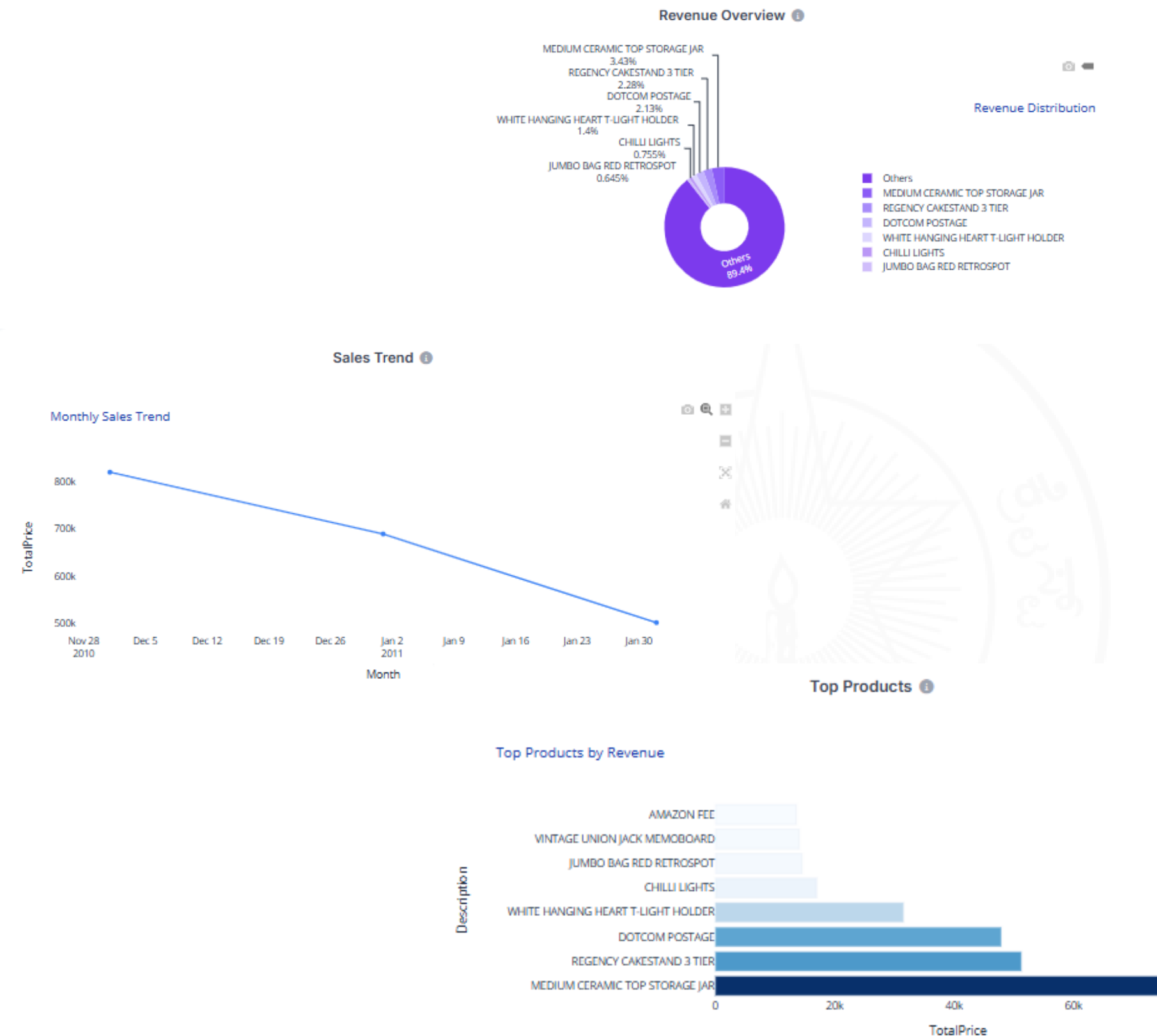
- Shows contribution of products to overall revenue.
- Useful for identifying top revenue drivers.

2. Monthly Sales Trend (Line Chart)

- Revenue plotted month by month.
- Helps detect growth, seasonality, or dips.

3. Top Products by Revenue (Horizontal Bar Chart)

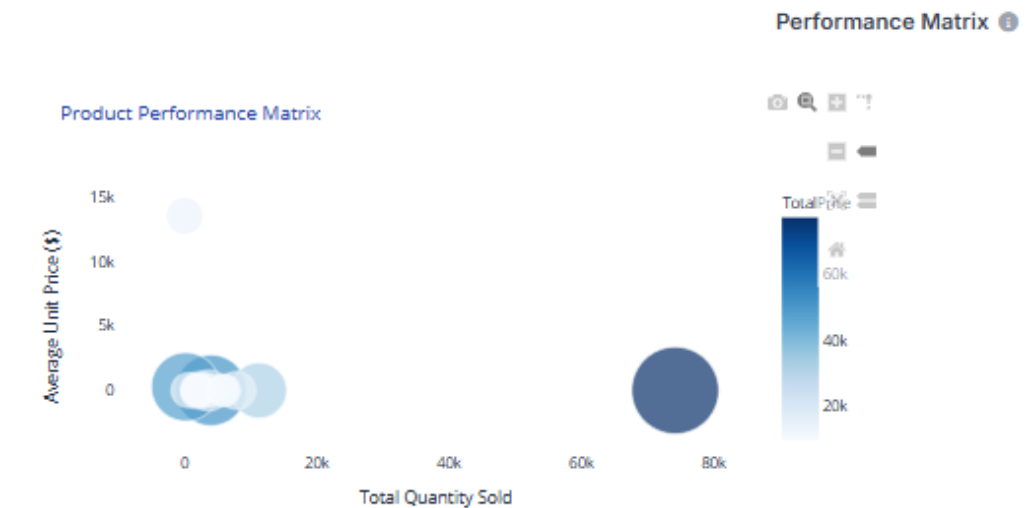
- Lists highest revenue-generating products.
- Retailers can prioritize inventory & promotions.



DASHBOARD - PRODUCT PERFORMANCE TAB

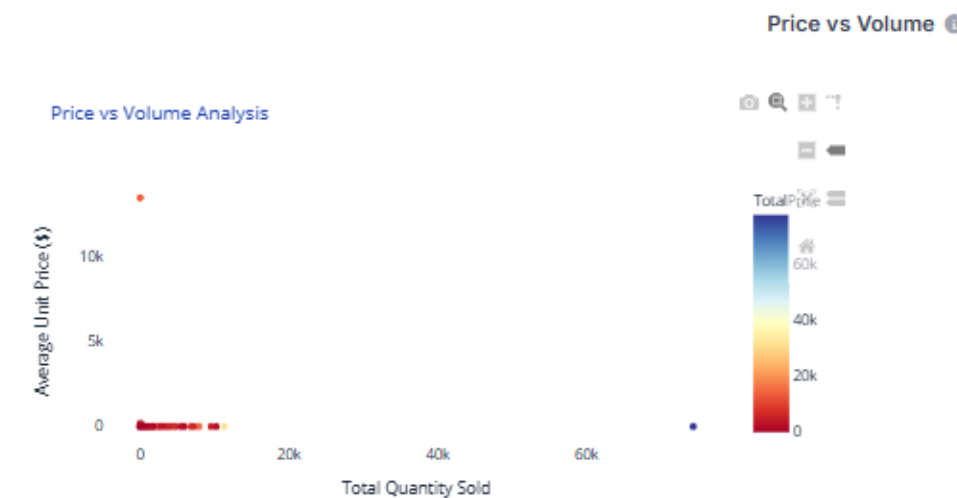
1. Product Performance Matrix (Bubble Chart)

- Plots products by sales volume, average price, and bubble size = revenue.
- Identifies high-volume, high-revenue, and premium products.



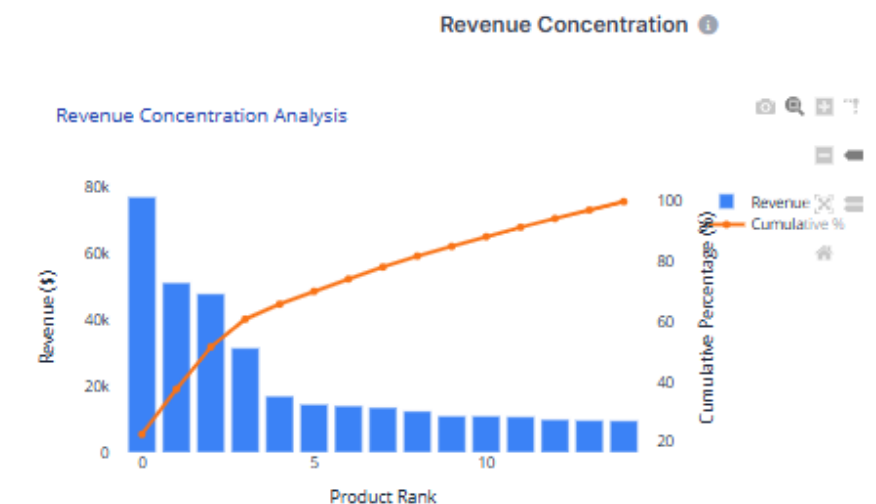
2. Price vs Volume Analysis (Scatter Plot)

- Compares unit price with sales volume.
- Useful to see demand sensitivity to pricing.



3. Revenue Concentration (Pareto Chart)

- Combines bar (revenue by product) + line (cumulative %).
- Reveals “80/20” products driving majority of sales.



DASHBOARD - PRODUCT PERFORMANCE TAB

4. Performance by Price Category (Grouped Bar Chart)

- Groups items into Budget, Mid, Premium, Luxury.
- Shows how categories contribute to quantity & revenue.



5. Volume Distribution by Product (Box Plot)

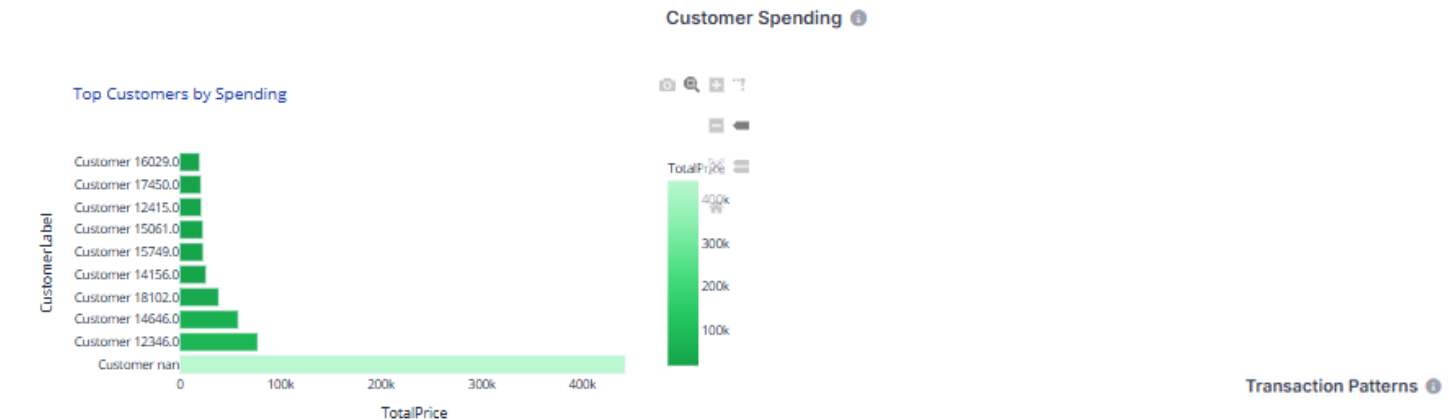
- Shows spread of sales volume across top products.
- Highlights outliers and consistency of demand



DASHBOARD - CUSTOMER INSIGHTS TAB

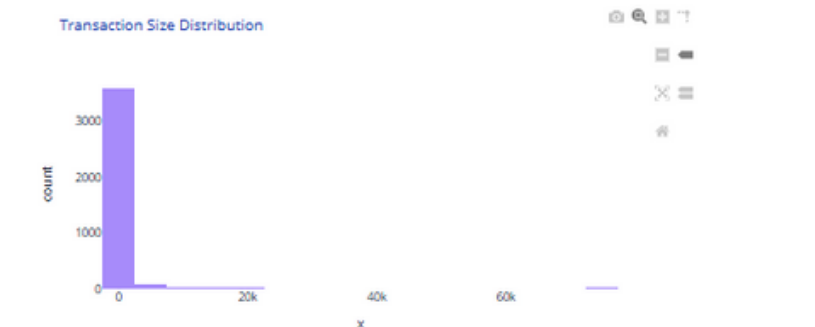
1. Top Customers by Spending (Bar Chart)

- Shows customers who generate the highest revenue.
- Helps in loyalty & targeted offers.



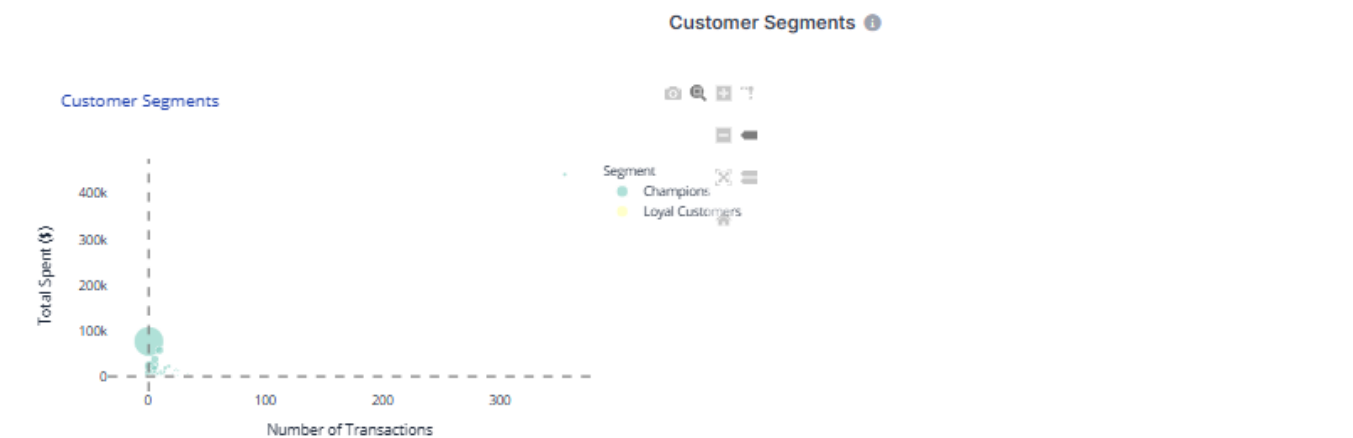
2. Transaction Size Distribution (Histogram)

- Plots frequency of transaction values.
- Shows typical customer spend range.



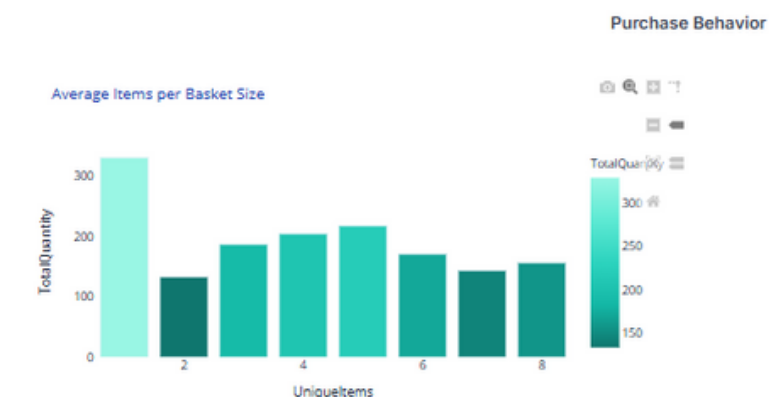
3. Customer Segments (Scatter Plot with Clusters)

- Segments: Champions, Loyal, Big Spenders, New Customers.
- Based on spending & purchase frequency.



4. Purchase Behavior – Avg Items per Basket Size (Bar Chart)

- Avg quantities bought vs unique items per basket.
- Helps design bundles that match natural buying patterns.



DASHBOARD - ADVANCED ANALYTICS TAB

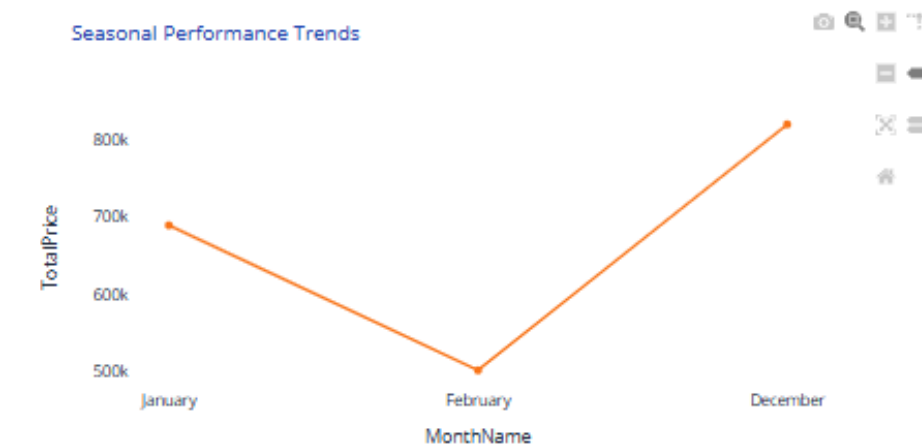
1. Cross-Sell Heatmap (Matrix)

- Shows which products are often bought together.
- Ideal for bundle or “frequently bought together” offers.



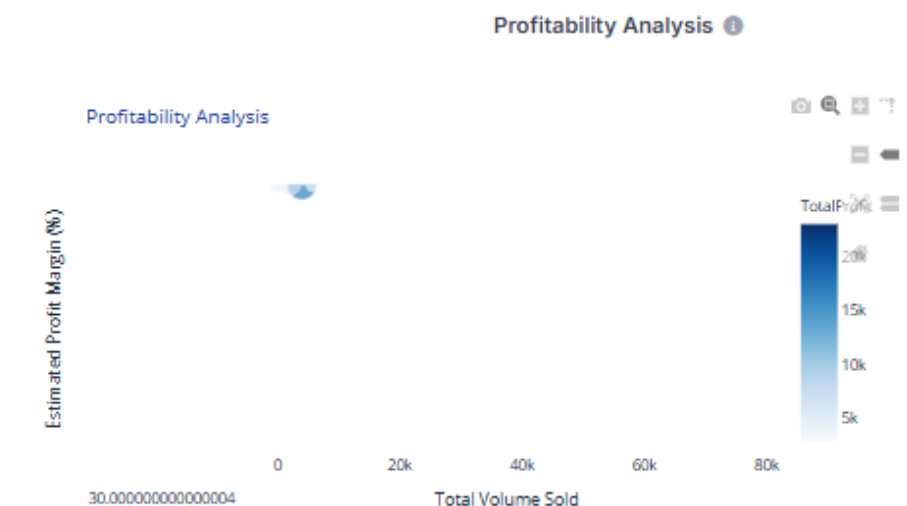
2. Seasonal Performance Trends (Line Chart)

- Monthly revenue trend across the year.
- Retailers can plan festive/seasonal promotions.



3. Profitability Analysis (Scatter/Bubble Chart)

- Plots products by sales volume vs profit margin, bubble size = total profit.
- Identifies high-margin vs low-margin products.



RESULTS ACHIEVED

- **Data Handling** → Upload in CSV, XLSX, PDF + automated preprocessing for clean datasets
- **Bundle Intelligence** → Identified complementary & sequential bundles using robust algorithms
- **Pricing Optimization** → Applied strategies delivering higher margins & revenue impact
- **Interactive Dashboard** → 15+ visualizations for revenue, customers, products, and trends
- **User-Friendly UI** → Simple navigation with export options (Excel/PDF)
- **Retailer Benefits** → Faster decisions, improved bundling, targeted promotions, measurable revenue uplift

FUTURE SCOPE

Integration with Databases

- Add support for SQL/NoSQL databases to handle larger datasets and enable persistent storage.

Advanced Scalability

- Implement distributed computing techniques (e.g., Apache Spark) for faster MBA/SPM analysis on enterprise-scale data.

Enhanced User Management

- Introduce role-based access control (RBAC) to support multiple users with varying privileges.

Automated Parameter Tuning

- Enable the system to dynamically suggest or adjust MBA/SPM parameters (support, confidence, lift thresholds) based on dataset characteristics

Cloud Deployment

- Deploy BundleWise on cloud platforms (AWS, Azure, GCP) to increase accessibility and ensure availability for large-scale usage.

Integration with POS Systems

- Directly connect with Point-of-Sale (POS) systems for real-time ingestion and near real-time analytics.

Improved Reporting Features:

- Allow customizable report templates, with more flexibility in selecting which KPIs and charts are included in exports.



CONCLUSION

BundleWise has been designed as an end-to-end solution that streamlines the entire retail analytics process. From multi-format data upload and automated preprocessing to advanced bundle identification and optimized pricing strategies, the system integrates robust analytics with a simple, intuitive dashboard. Retailers can easily explore sales, customer behavior, and product performance through interactive visualizations, gaining actionable insights without requiring technical expertise. By enabling smarter bundle recommendations, effective cross-sell opportunities, and profitability-focused pricing, BundleWise not only simplifies decision-making but also drives measurable business growth. This platform ultimately bridges raw data with practical retail strategies for sustained competitive advantage.

Thank You



